Geometric Group Theory

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Lecture 1

1 Combinatorial group theory

Combinatorial group theory is a sibling field to Geometric group theory. Both study infinite discrete groups.

1.1 Free groups and presentations

Let $A = \{a_1, a_2, \dots\}$ be an *alphabet*. A group F is *free on* A if

1. There is a map of sets $A \to F$,
2. for any group G, and a map of sets $A \to G$, there exists a unique group homomorphism $F \to G$ such that

$$\begin{array}{ccc} & & F \\ & \nearrow & \downarrow \\ A & \longrightarrow & G \end{array}$$

commutes.

This is a *universal property*. As usual, this means that F is unique up to unique isomorphism. This shows that F is determined by A, so we may write $F = F(A)$.

However (as usual with definitions by universal property), we don't know if $F(A)$ exists. We'll show this two different ways.

1. Topologically: Let

$$X = \bigvee_{a \in A} S^1$$

By the Seifert-van Kempen theorem,

$$\pi_1(X) \cong F(A)$$

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2. **Combinatorially:** Let

$$A^* = \{\text{words in } A \sqcup A^{-1}\}$$

where $A^{-1} = \{a_1^{-1}, \dots, a_n^{-1}\}$. For example,

$$1 = \varnothing, aa, aa^{-1}, aba^{-1}b^{-1}, a^{100}ba^{-100}b, \dots$$

A word is *reducible* if it contains aa^{-1} or $a^{-1}a$ as a subword for any $a \in A$. Otherwise, it is *reduced*. We can now define

$$F(A) = \{w \in A^* \text{ reduced}\}$$

The group operation is concatenation, followed by reduction. For example,

$$(abab^{-1})(b^2a) = abab^{-1}b^2a = ababa$$

Note that reduction terminates as each reduction step reduces the length. We won't check that this is well defined or associative. The identity element is 1, inverses is clear.

A *presentation* consists of an alphabet A, which we will call *generators*, and a set $R \subseteq F(A)$, which we will call *relations*, and we write

$$\langle A \mid R \rangle = \langle a_1, a_2, \cdots \mid r_1, r_2, \dots \rangle$$

This *presents* a group

$$G = \frac{F(A)}{\langle\langle R \rangle\rangle}$$

where $\langle\langle R \rangle\rangle$ is the *normal closure* of R, i.e. the smallest normal subgroup of $F(A)$ containing R.

Example 1.1.1

Some examples of presentations which we have seen:

$$\begin{aligned}
\langle a \mid a^n \rangle &\cong \mathbb{Z}/n\mathbb{Z} \\
\langle r, s \mid r^n, s^2, srsr \rangle &\cong D_{2n} \\
\langle A \mid \rangle &\cong F(A) \\
\langle a, b \mid aba^{-1}b^{-1} \rangle &\cong \mathbb{Z}^2 \\
\langle a_1, \dots, a_g, b_1, \dots, b_g \mid a_1b_1a_1^{-1}b_1^{-1} \cdots a_gb_ga_g^{-1}b_g^{-1} \rangle &\cong \pi_1(\Sigma_g)
\end{aligned}$$

where Σ_g is the compact orientable surface of genus g.

As we see, presentations arise when we write down fundamental groups of spaces. In fact, all groups arise this way.

Corollary 1.1.2 (of Seifert-van Kampen)**.** For

$$G = \langle a_1, a_2, \cdots \mid r_1, r_2, \dots \rangle$$

there exists a space X with $\pi_1(X) = G$.

Proof. First, start of with a wedge of circles, one for each a_i. Also consider a disjoint union of discs, one for each r_i. Attach the i-cell along its boundary, which is a loop in the wedge of circles. □

This is called the *presentation complex* of G. For example, if we have $\mathbb{Z}^2 = \langle a, b \mid aba^{-1}b^{-1} \rangle$, then $X = \mathrm{T}^2$ is the 2-torus.

In 1911, Max Dehn posed the following problems.

1. (Word problem) Given $w \in A^*$, determine whether or not $w = 1$ in $G = \langle A \mid R \rangle$. That is, whether or not $w \in \langle\langle R \rangle\rangle \subseteq F(A)$.
2. (Conjugacy problem) Given $G = \langle A \mid R \rangle$, $u, v \in A^*$, determine whether or not u is conjugate to v in G.
3. (Isomorphism problem) Given $G = \langle A \mid R \rangle$, $H = \langle B \mid S \rangle$, determine whether or not G is isomorphic to H.

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Remark 1.1.3. The conjugacy problem is stronger than the word problem, since $w = 1$ if and only if w is conjugate to 1.

Dehn was motivated by topology, but the problems asks for algorithms. We will often solve them using geometry.

All three were *un*solved in the 1950s, as all three problems are algorithmically undecidable. Nevertheless, positive solutions are known for many reasonable classes of groups.

Let $A = \{a_1, \dots, a_n\}$ be a finite alphabet.

Example 1.1.4 (word problem in free groups)

Let $w \in A^*$. If w is reduced, then $w = 1$ if and only if $w = \varnothing$. Otherwise, w contains a subword aa^{-1} for some $a \in A \cup A^{-1}$. Cancelling aa^{-1} produces a word $w' \in A^*$, such that $w = w'$ in $F(A)$, and $\ell(w') = \ell(w) - 2 < \ell(w)$. This terminates after finitely many steps.

We can also solve the conjugacy problem for free groups.

Definition 1.1.5

There is a natural action of $\mathbb{Z}$ on A^* permuting words. That is,

$$1 \cdot a_1 \cdots a_k = a_2 a_3 \cdots a_k a_1$$

The elements of $\mathbb{Z} \cdot w$ are called the *cyclic conjugates* of w. Note that all $u \in \mathbb{Z}w$ are conjugate to w. The orbits $\mathbb{Z}\backslash A^*$ are called *cyclic words*. A word is *cyclically reduced* if every cyclic conjugate is reduced.

Example 1.1.6

aba^{-1} is reduced, but not cyclically reduced as $ba^{-1}a$ is not reduced.

Note that if w is reduced, but not cyclically reduced, then

$$w = aw'a^{-1}$$

for some $a \in A \cup A^{-1}$. Note that w' is both conjugate to w and shorter than w. Hence after finitely many iterations, we can assume that w is cyclically reduced.

Lemma 1.1.7 (conjugacy problem in free groups)**.** If $u, v \in F(A)$ are cyclically reduced, then u is conjugate to v if and only if the corresponding cyclic words are the same.

Proof. If u, v have the same cyclic words, then v is a cyclic conjugate of u, which we have seen is a conjugate of u.

Conversely, suppose $u = gvg^{-1}$. By induction on $\ell(g)$, we can assume $g = a \in A \cup A^{-1}$. From this it follows that either $v = a^{-1}v'$, or $v = v'a$, as v is cyclically reduced. That is, $u = v'a^{-1}$ or $u = av'$. In both cases, they are cyclic conjugates. □

1.2 Historical case study

Let's briefly think about the state of topology in the early 20th century. Poincaré knew that homology classifies the compact two-dimensional surfaces. This motivated the

Conjecture 1.2.1 (Poincaré conjecture, version 1)**.** Let M be a compact connected 3-manifold, with

$$\mathrm{H}_*(M) = \begin{cases} \mathbb{Z} & * = 0, 3 \\ 0 & \text{otherwise} \end{cases}$$

Then M is homeomorphic to S^3.

Such a 3-manifold M is called a *homology sphere*.

Theorem 1.2.2 (Poincaré, 1904)**.** There exists a three dimensional homology sphere P, with

$$\pi_1(P) \twoheadrightarrow A_5$$

The moral is that: homology is not enough, we need use π_1 as well.

Conjecture 1.2.3 (Poincaré conjecture, version 2)**.** Let M be a compact connected 3-manifold, with $\pi_1(M) = 1$. Then M is homeomorphic to S^3.

This was proven by Perelman in 2003. Returning to the original conjecture, in 1910 Dehn wanted to construct more homology spheres.

Theorem 1.2.4 (Dehn, 1910)**.** There are infinitely many non-homeomorphic 3-dimensional homology spheres.

Remark 1.2.5. The isomorphism problem is exactly what is needed to distinguish these manifolds.

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Here is Dehn's construction. Consider the trefoil knot $K \subseteq \mathbb{R}^3 \subseteq S^3$. Let $N(K)$ be a regular open neighbourhood of K, i.e. a thickening of K. Let $N = S^3 \setminus N(K)$. This is a compact 3-manifold with boundart $\partial N \cong \mathrm{T}^2$.

Computing,

$$\mathrm{H}_*(N) = \begin{cases} \mathbb{Z} & * = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$\pi_1(N) \cong \langle x, y, z \mid x^2 = y^3 = z \rangle$$

This follows from the fact that K is a torus knot, and we can compute the π_1 of the complement of a torus knot. Note that the abelianisation map

$$\pi_1(N) \to \mathrm{H}_1(N) \cong \mathbb{Z}$$

sends z to 6, x to 3 and y to 2. Moreover, the boundary torus T has $\pi_1(T) = \mathbb{Z}^2$, generated by xy and z. Under the abelianisation map, xy is sent to 5.

We can glue a solid torus $D^2 \times S^1$ to N, by a homeomorphism on the boundary. Let λ, μ be the corresponding loops on T^2. The resulting manifold $M_\phi = N \cup_\phi T$ is closed. $\pi_1(M_\phi)$ depends on $g = \phi_*(\mu)$. By Seifert-van Kapmen,

$$\pi_1(M_\phi) = \frac{\pi_1(N)}{\langle\langle g \rangle\rangle}$$

Similarly,

$$\mathrm{H}_1(M_\phi) = \frac{\mathbb{Z}}{\langle [g] \in \mathrm{H}^1(N) \rangle}$$

To produce a homology sphere, we need to choose ϕ such that $g = \phi_*(\mu) = 1$ in $\mathrm{H}^1(N)$.

If $g = (xy)^a z^b$, then in $\mathrm{H}^1(N)$ this is mapped to $5a + 6b \in \mathbb{Z}$. Choose $a = 6n + 5$, and $b = -(5n + 4)$ for $n \in \mathbb{Z}$. He constructs ϕ_n such that

$$\phi_n(\mu) = g_n = (xy)^{6n+5} z^{-(5n+4)}$$

His family of manifolds

$$D_n = N \cup_{\phi_n} U$$

has

$$\pi_1(D_n) = \langle x, y, z \mid x^2 = y^2 = z, (xy)^{6n+5} = z^{5n+4} \rangle$$

The remaining challenge is to prove the groups $G_n = \pi_1(D_n)$ for $n \geq 0$ are pairwise non-isomorphic.

This is the isomorphism problem! In particular,

$$g_n = g_m \implies g_n \text{ and } g_m \text{ are conjugate} \implies G_n \cong G_m$$

So we also need to solve the word and conjugacy problem in $\pi_1(N)$.

1.3 van Kampen diagrams

Definition 1.3.1

A map $f : Y \to X$ of cell complexes is called *combinatorial* if for all $k \in \mathbb{Z}_{\geq 0}$, and every k-cell e^k of Y, f maps the interior $\mathrm{Int}(e^k)$ homeomorphically to the interior of a k-cell of X.

Consider a presentation $G = \langle a_i \mid r_j \rangle$, and the associated presentation complex X.

Definition 1.3.2

A *(singular) disc diagram* is a compact contractible 2-complex D, with an embedding $D \hookrightarrow \mathbb{R}^2$. A disc diagram D is *over* X if it is equipped with a combinatorial map $D \to X$.

Example 1.3.3

Example of disc diagram.

Recall that X is given by a wedge of circles, with discs glued on for each relation. So the 1-cells correspond to generators, 2-cells go to relations (or cyclic conjugates, or inverses). With this:

- every oriented 1-cell of D is labelled with some $a_i \in A$,
- so that each 2-cell has boundary which is a cyclic conjugate of some $r_j^{\pm 1}$.

Associated to each disc diagram D, we have a *boundary cycle*, which reads a (cyclic[1]) word $w \in A^*$, which reduces to an element $w' \in \langle\langle r_j \rangle\rangle \leq F(A)$. To see this, D is contractible.

D is a *van Kampen diagram for* w.

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Lemma 1.3.4 (van Kampen)**.** If $w \in \langle\langle R \rangle\rangle$, then there exists a van Kampen diagram for w.

Proof. Suppose $w \in \langle\langle R \rangle\rangle$. Then w can be written as

$$w = \prod_{i=1}^{k} h_i r_i^{\pm 1} h_i$$

in $F(A)$, where $h_i \in F(A)$, and $r_i \in R$. Now build a *lollipop diagram* D_0, which has boundary word w_0, which is equal to w in $F(A)$, but may not be reduced.

If w_0 is reduced, $w = w_0$, and so we are done. Otherwise, w_0 contains a cancelling pair, so

$$w_0 = \cdots a a^{-1} \cdots$$

for some $a \in A \cup A^{-1}$. In w_0, this looks like:

We can see that e_1, e_2 share a vertex. There are two cases to consider:

1. if the *origin* of e_1 is the *terminus* of e_2, then the diagram D_0 is a wedge $D_1 \vee D'$. Then D_1 is a van Kampen diagram for w_1, which is the result of cancelling a and a^{-1}.
2. if the origin of e_1 is distinct from the terminus of e_2, then we can fold the edges to get D_1, which has boundary word w_1 as above.

In either case, $w_1 = \partial D_1$ is obtained from w_0 by cancelling a pair. Therefore, we may proceeed by induction, and after finitely many repetitions, we construct a van Kampen diagram D_n such that $w_n = \partial D_n$ is reduced, and $w_n = w$ in $F(A)$. Thus, $w_n = w$ as words, and so D_n is a van Kampen diagram for w. □

[1] or a word once we choose a base point.

Remark 1.3.5. The minimal number of 2-cells in a van Kampen diagram of w is the minimal number of k, such that w can be written as

$$w = \prod_{i=1}^{k} h_i r_i^{\pm 1} h_i^{-1}$$

This is called the *area* of w.

Example 1.3.6

Let $G = \mathbb{Z}^2 = \langle a, b \mid [a, b] \rangle$. Consider $w = a^n b^n a^{-n} b^{-n}$. This has van Kampen diagram

In this case, $\text{Area}(D) = n^2$. We will show D is minimal, and so $\text{Area}(w) = n^2$.

Definition 1.3.7

Let $\mathcal{P} = \langle A \mid R \rangle$ be a finite presentation of a group G. Define

$$\begin{aligned} \delta_{\mathcal{P}} : \mathbb{N} &\to \mathbb{N} \\ \ell &\mapsto \max_{w \in \langle\langle R \rangle\rangle, \ell(w) = d} \text{Area}(w) \end{aligned}$$

This is called the *Dehn function*.

Remark 1.3.8. The word problem in $\mathcal{P}$ if and only if $\delta_{\mathcal{P}}$ is computable.

2 Basics of geometric group theory

2.1 Cayley graphs

A *graph* is a 1-dimensional cell complex. Throughout, let G be a group, with finite generating set $S \subseteq G$.

Definition 2.1.1

The *Cayley graph* $\text{Cay}_S(G)$ is defined as follows:

- vertices $V(\text{Cay}_S(G)) = G$,
- edges $E(\text{Cay}_S(G))$ correspond bijectively with $G \times S$. That is, we have an edge $g \to gs$.

Example 2.1.2

The trivial group given by $1 = \langle a, b \mid a, b \rangle$ has Cayley graph

Example 2.1.3

$S_3 = \langle r, s \mid srsr, r^3, s^2 \rangle$ has Cayley graph

Example 2.1.4

$\mathbb{Z} = \langle 1 \rangle$ as Cayley graph

On the other hand, $\mathbb{Z} = \langle 2, 3 \rangle$ has Cayley graph

Note that the action of G on itself on the *left* extends to an action of G on $\text{Cay}_S(G)$, sending an edge $h \to hs$ to $gh \to ghs$. Note the right action does not work, because of our definition of the Cayley graph.

Remark 2.1.5. The action of G on $\mathrm{Cay}_S(G)$ is free. That is, for all $x \in \mathrm{Cay}_S(G)$, $\mathrm{Stab}_G(x) = 1$.

Proposition 2.1.6. Let $G = \langle S \mid R \rangle$, and let X be the corresponding presentation complex. Then there exists a G-equivariant isomorphism of graphs

$$\mathrm{Cay}_S(G) \cong \widetilde{X}_{(1)}$$

with the 1-skeleton of the universal cover $\widetilde{X}$ of X[a].

[a] and not the other way around.

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Proof. Consider the natural free action of $G = \pi_1(X)$ on $\widetilde{X}$, by deck transformations. The action is by combinatorial endomorphisms. In particular, it preserves the 1-skeleton. So we have a free action of G on $\widetilde{X}_{(1)}$, which sends vertices to vertices and edges to edges.

The action of G on $\widetilde{X}_{(0)}$ is free, and as X has only one vertex, transitive. Therefore, choosing a base vertex $\widetilde{v}_0$, the orbit-stabiliser theorem provides a G-equivariant bijection

$$G \to \mathrm{Orb}(\widetilde{v}_0) = \widetilde{X}_{(0)}$$

sending g to $g \cdot \widetilde{v}_0$. So this matches up the vertices as required.

Next, let us match up the edges. For each $s \in S$, let e_s be the corresponding edge of X. Let $\widetilde{e}_s$ be the unique lift of e_s to $\widetilde{X}$, beginning at $\widetilde{v}_0$. By the definition of the action of G on $\widetilde{X}$, e_s ends at $s \cdot \widetilde{v}_0$. Now an arbitrary edge $\widetilde{e}$ of $\widetilde{X}$ maps to some e_s, under the covering map. Since egdes of X correspond to G-orbits of edges in $\widetilde{X}$, it folloes that $\widetilde{e} = g \cdot \widetilde{e}_s$ for some $s \in S$. That is, $\widetilde{e}$ is the edge from $g \cdot \widetilde{v}_0$ to $gs \cdot \widetilde{v}_0$. So it corresponds to an edge from g to gs.

This shows that the G-equivariant map $G \to \widetilde{X}_{(0)}$ extends to a G-equivariant isomorphism of graphs as claimed. □

The next proposition deepens the relationship between generating sets and path connectedness.

Proposition 2.1.7. Let $\widetilde{X}$ be a path connected topological space, and suppose that G acts on $\widetilde{X}$ by homeomorphisms. If $U \subseteq \widetilde{X}$ is an open subset, such that $G \cdot U = \widetilde{X}$, then the set

$$S = \{g \in G \mid g \cdot U \cap U \neq \varnothing\}$$

generates G.

Proof. Fix a base point $\widetilde{x}_0 \in U$. Now for $g \in G$, let $\gamma : [0,1] \to \widetilde{X}$ be a path from $\widetilde{x}_0$ to $g \cdot \widetilde{x}_0$. The set $\{\gamma^{-1}(h \cdot U) \mid h \in G\}$ is an open cover of $[0,1]$. So it has a finite subcover, $\{\gamma^{-1}(U_1), \ldots, \gamma^{-1}(U_n)\}$, where $U_i = g_i \cdot U$. We may choose the indices so that

- $\widetilde{x}_0 \in U_1$,
- $\gamma^{-1}(U_i) \cap \gamma^{-1}(U_{i+1}) \neq \varnothing$ for all i,
- $g \cdot \widetilde{x}_0 \in U_n$.

Note that the g_i need not be unique. By definition, $x_0 \in U \cap g_1 \cdot U$, and so $g_1 \in S$. Similarly, if $t_i \in \gamma^{-1}(U_i) \cap \gamma^{-1}(U_{i+1})$, then $x_i = \gamma(t_i) \in g_i \cdot U \cap g_{i+1} \cdot U$. Thus,

$$g_i^{-1} \cdot x_i \in U \cap g_i^{-1} g_{i+1} \cdot U$$

and so $s_i = g_i^{-1} g_{i+1} \in S$. Thus, $g_n = s_{n-1} \cdots s_2 s_1 g_1$, is a finite product of elements of S. Finally, $g^{-1} g_n \in S$ similarly to the above, so $g \in \langle S \rangle$ as required. □

Example 2.1.8

Let $\Gamma \subseteq \mathrm{Isom}(\mathbb{R}^2)$ be the symmetry group of the standard tiling of the plane by equilateral triangles. Let U be a thickened triangle. Using the proposition, we obtain a finite generating set of Γ. In particular, Γ is generated by the reflections in the sides of a single triangle.

In particular, this is not a covering space action, as it is not free.

Definition 2.1.9

An action of G on $\widetilde{X}$ by homeomorphisms is *properly discontinuous* if for every compact $K \subseteq \widetilde{X}$, the set

$$\{g \cdot K \cap K\}$$

is finite.

The action is *cocompact* if there exists $K \subseteq \widetilde{X}$ compact, such that

$$G \cdot K = \widetilde{X}$$

$\widetilde{X}$ is *locally compact* if for every neighbourhood U of x, there exists an open neighbourhood $V \subseteq U$ of x, such that $\overline{V} \subseteq U$ is compact.

Corollary 2.1.10. If G acts on $\widetilde{X}$ properly discontinuously and cocompactly, and $\widetilde{X}$ is path connected and locally compact, then G is finitely generated.

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Proof. Let $K \subseteq \widetilde{X}$ be compact, such that $G \cdot K = \widetilde{X}$. By local compactness, we may find an open U such that $K \subseteq U$, and $\overline{U}$ is compact. In particular, $G \cdot U = \widetilde{X}$, and the set

$$S = \{g \in G \mid g \cdot U \cap U \neq \varnothing\} \subseteq \{g \mid g \cdot \overline{U} \cap \overline{U} \neq \varnothing\}$$

But the right hand side is finite, so S is finite. By the proposition, S generates G. □

Corollary 2.1.11. If X is compact, locally compact and has a universal cover $\widetilde{X}$, then $\pi_1(X)$ is finitely generated.

Proof. Exercise. Sheet 1 question 10. □

2.2 The Schwarz-Milnor lemma

Cayley graphs are not just combinatorial. They admit a natural metric, called the *word metric*.

Definition 2.2.1 (word metric)

Let S generate G. Define

$$\ell_S(G) = \min\{n \mid g = \prod_{i=1}^{n} s_i^{\pm 1}, s_i \in S\}$$

This defines a metric

$$d_S(g, h) = \ell_S(g^{-1}h)$$

called the *word metric* associated to S.

The word metric is invariant under the *left* G action on itself. That is,

$$d_S(\gamma g, \gamma h) = d_S(g, h)$$

However, it is, in general, not right invariant.

Example 2.2.2

$G = \mathbb{Z}^2 = \langle a \rangle \oplus \langle b \rangle$. Then the word metric is just the ℓ_1-metric.

Remark 2.2.3. The word metric extends naturally to a left invariant metric on $\mathrm{Cay}_S(G)$, in which the interior of each edge is locally isometric to $(0, 1)$. That is, the path metric.

Lemma 2.2.4. Suppose S, T are finite generating sets for G. Then there exists constants $C, C' \geq 1$ such that

$$\frac{1}{C} d_T \leq d_S \leq C' d_T$$

Proof. Let $C = \max_{s \in S} \ell_T(s)$. Then for any $g \in G$,

$$\ell_T(g) \leq C \ell_S(g)$$

by induction. □

That is, for finitely generated groups, the word metric is well defined, up to bi-Lipschitz equivalence.

Definition 2.2.5 (quasi-isometry)

A function[a] $f : X \to Y$ between metric spaces is a *quasi-isometric embedding* if there are constants $C \leq 1$, $D \geq 0$, such that

$$\frac{1}{C} d(x, x') - D \leq d(f(x), f(x')) \leq C d(x, x') + D$$

for all $x, x' \in X$.

If in addition, there exists a constant K such that for every $y \in Y$, there exists $x \in X$ such that $d(f(x), y) \leq K$, then f is called a quasi-isometry, and we write $X \overset{\text{qi}}{\sim} Y$.

[a] It does not have to continuous.

Remark 2.2.6. On examples sheet 1, we have that $\overset{\text{qi}}{\sim}$ is an equivalence relation.

Example 2.2.7

Every bounded metric space is quasi-isometric to a point.

Definition 2.2.8 (proper)

A metric space X is *proper* if closed balls in X are compact.

Definition 2.2.9 (geodesic)

A *geodesic* in X is an isometric embedding $\gamma : [a, b] \to X$. The metric space X is *geodesic* if every pair of points is joined by a geodesic.

Theorem 2.2.10 (Schwarz-Milnor). Suppose X is a proper geodesic metric space. Let G acts on X properly discontinuously and cocompactly by isometries. Then G is finitely generated, and

$$X \overset{\text{qi}}{\sim} (G, d_S)$$

for any finite generating set S of G.

Proof. Fix a base point $x_0 \in X$. Let $B = \overline{B}(x_0, K) \subseteq X$ be a closed ball, such that $G \cdot B = X$. By properness and proper discontinuity, the set

$$\{g \in G \mid d(x_0, g(x_0)) \leq 3K\}$$

is finite. Therefore, there exists $\varepsilon > 0$, such that

$$d(x_0, g(x_0)) < 2K + \varepsilon \iff d(x_0, g(x_0)) \leq 2K$$

Moreover, in this case, $gB \cap B \neq \varnothing$.

Lecture 7

If $U = B(x_0, K + \varepsilon/2)$, then

$$S = \{g \cdot U \cap U \neq \varnothing\} = \{g \cdot B \cap B \neq \varnothing\}$$

Since B is compact, S is finite, since the action is properly discontinuous. But S is a generating set for G.

Since the word metric for any two finite generating sets are bi-Lipschitz, we may prove the result for the S above. Consider the map $f : G \to X$, $f(g) = g \cdot x_0$. We claim that this is a quasi-isometry. f is quasi-surjective, since $G \cdot B = X$. It remains to prove that f is a quasi-isometric embedding. That is, we want upper and lower bounds on $d(x_0, g \cdot x_0)$ in terms of $\ell_S(g)$.

For the upper bound, take $C = \max_{s \in S} d(x_0, s \cdot x_0)$. Then

$$d(x_0, g \cdot x_0) \leq C\ell_S(g)$$

for any $g \in G$, using the triangle inequality.

For the lower bound, consider a geodesic $\gamma : [0, d(x_0, g \cdot x_0)] \to X$ from x_0 to $g \cdot x_0$. Choose a dissection of $[0, d(x_0, g \cdot x_0)]$

$$0 = t_0 < t_1 < \cdots < t_n < t_{n+1} = d(x_0, g \cdot x_0)$$

with

$$\frac{\varepsilon}{2} \leq |t_i - t_{i+1}| < \varepsilon$$

for $0 \leq i \leq n - 1$. Note we can make it so that $|t_n - t_{n+1}| < \varepsilon$, but we may not have the lower bound.

Since $G \cdot B$ covers X, for $1 \leq i \leq n$, there exists $g_i \in G$ such that $\gamma(t_i) \in g_i \cdot B$. Set $g_0 = 1$ and $g_{n+1} = g$. Then $\gamma(t_i) \in g_i \cdot B$ for all i. For each i,

$$d(g_i(x_0), g_{i+1}(x_0)) < 2K + \varepsilon$$

by the triangle inequality. Therefore, $g_i^{-1} g_{i+1} \in \{h \cdot U \cap U \neq \varnothing\} = S$. Hence $\ell_S(g) \leq n + 1$. Furthermore,

$$|t_i - t_{i-1}| \geq \varepsilon/2$$

for all $1 \leq i \leq n$, so $d(x_0, g \cdot x_0) \geq n\varepsilon/2$. Combining these,

$$\ell_S(g) \leq n + 1 \leq \frac{2}{\varepsilon} d(x_0, g \cdot x_0) + 1$$

We can rearrange this to get the lower bound. □

Example 2.2.11

Recall the two Cayley graphs of $\mathbb{Z}$, with generating sets $\{1\}$, and $\{2, 3\}$ respectively.

diagram

The Schwartz-Milnor says that these are both quasi-isometric to $\mathbb{Z}$ with an appropriate word metric. So they are quasi-isometric. More generally, for any finitely generated group G, the Cayley graphs of any two finite generating sets are quasi-isometric.

Corollary 2.2.12. If G is finitely generated, H is a subgroup with finite index in G. Then H is finitely generated, and H is quasi-isometric to G.

Proof. H acts on $\mathrm{Cay}_S(G)$. The action is cocompact as H has finite index. So it satisfies the Schwartz-Milnor lemma. □

Example 2.2.13

Let Σ_2 be the closed orientable surface of genus 2, and $G = \pi_1(\Sigma_2)$. Choose a Riemannian metric g on Σ_2 of constant curvature -1.

This pulls back to a Riemannian metric on its universal cover $\widetilde{\Sigma}_2$. By a classical theorem of differential geometry, $\widetilde{\Sigma}_2$ is isometric to the hyperbolic plane $\mathbb{H}^2$. Moreover, the action of G on the $\mathbb{H}^2$ by isometries, and properly discontinuously. The action is cocompact as the quotient is Σ_2, which is compact. So by the Schwartz-Milnor lemma, $\pi_1(\Sigma_2)$ is quasi-isometric to $\mathbb{H}^2$.

3 Case study - Free groups

Let $A = \{a_1, \dots, a_n\}$. We will write $F_n = F(A_n)$. The *Cayley tree* is the infinite $2n$-valent tree $T_n = \mathrm{Cay}_A(F_n)$.

diagram for $n = 2$

Lecture 8

In particular, every vertex looks the same. F_n acts freely on T_n. The quotient X_n is the wedge of n circles. So we recover $F_n = \pi_1(X_n)$, and T_n is the universal cover of X_n.

We can translate our *combinatorial* arguments about F_n, about geometric properties of T_n.

Words

A word $w \in A^*$ is equivalent to an *edge path*, which is a map $w : I \to X_n$, where I is an interval.

For example, consider the word $w = a^2bb^{-1}a^{-1}$. The edge path is: **diagram**

An edge path in X_n lifts to a unique edge path in T_n, based at 1. Conversely, each such path in T_n projects to a path in X_n.

diagram

Reduced words

A word $w \in A^*$ is reduced if and only if the corresponding edge path $w : I \to X_n$ is locally injective. In turn, this holds if and only if the corresponding edge path $w : I \to T_n$ is locally injective. This is because an edge path can only fail to be locally injective at a vertex.

Clearly, the shortest path in T_n from 1 to $g \in F_n$ is injective. In particular, locally injective. So every element of F_n is represented by a reduced word.

The fact that this representative is unique follows from the next lemma.

Lemma 3.0.1. If T is a tree, and $\gamma : I \to T$ is a locally injective (edge) path, then γ is injective.

Proof. Suppose not. Let $\gamma : [a, b] \to T$ be the shortest counterexample. In particular, $\gamma(a) = \gamma(b)$, and γ is injective on (a, b). So γ descends to an injective map $S^1 \to T$. But T is a tree. Contradiction. □

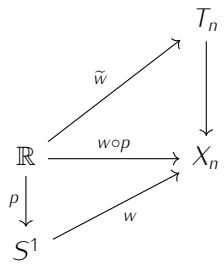
Similarly, if $g \in F_n$ is shortest such that g is represented by distinct reduced words w_1, w_2, then we get an embedding $S^1 \hookrightarrow T$. Hence the reduced word is unique.

For $g \in F_n$, write $[1, g]$ for the unique injective edge path from 1 to g.

Cyclically reduced words

So far, implicitly we have chosen base points. Each (nontrivial) word $w \in A^*$ also defines a (based) edge loop, by gluing together the end points of the interval. So we have a map $S^1 \to X_n$. If we forget the base point of S^1, then two elements $u, v \in A^*$ determine the same cyclic word if and only if they represent the same edge loop $S^1 \to X_n$.

Now a word $w \in A^*$ is cyclically reduced if and only if the corresponding map $S^1 \to X_n$ is locally injective. From lifting theory, we have a lift $\widetilde{w} : \mathbb{R} \to T_n$ as below



For example, if $w = ab^2$. the lift is

diagram

In particular, since w is locally injective, $\widetilde{w}$ is as well, and so it is injective, by the lemma. The image of $\widetilde{w}$ is called the *axis* of w.

By the definition of the action of F_n on T_n, w when thought of as a deck tranformation of T_n, preserves its axis. Note that w translates Axis(w) by $\ell(w)$. This is called the *translation length of* w, denoted as $\tau(w)$.

Lecture 9

A geometric solution to the conjugacy problem follows from:

Lemma 3.0.2. Let $u, v \in F_n$ be cyclically reduced. If u and v are conjugate, then there exists $g \in F_n$, such that

$$\ell(g) \leq \frac{1}{2}(\tau(u) + \tau(v))$$

and $u = gvg^{-1}$.

The conjugacy problem follows, as the lemma tells us that we only need to check $u = gvg^{-1}$ for finitely many g, and each of these can be checked using the word problem.

Remark 3.0.3. The statement is existence, it does not hold for all choice of g. In particular, $C(v)$ is infinite, as it contains v^k for all $k \in \mathbb{Z}$, and the length of gv^k is unbounded as $k \to \infty$.

In fact, the set of conjugators is the double coset $\langle u \rangle\, g\, \langle v \rangle$.

Proof. Suppose $u = gvg^{-1}$, with $\ell(g)$ minimal. Then

(i) If $u \in [1, g]$, then $g = uh$ for some h, and there is no cancellation. Moreover, $u = hvh^{-1}$, and if $h \neq g$, then $\ell(h) < \ell(g)$. Contradiction.

(ii) If $v \in [1, g^{-1}]$ is strictly between 1 and g as above, then $\ell(g)$ wasn't minimal.

Now consider the convex hull of $\{1, g, u, gv\}$.

diagram

For this, there are three (non-degenerate) different combinatorial types for the convex hull. The first case is

diagram

By the minimality in (i), $\ell(\alpha) > 0$. Similarly, $\ell(\beta) > 0$. On this diagram, we have Axis(u) and $g \cdot \text{Axis}(v) = \text{Axis}(gvg^{-1})$. But $u = gvg^{-1}$. Contradiction (we will assume the middle length is non-zero for now).

The second case is

diagram

The axes are labelled. But they translate in opposite directions. Contradiction (again, we assume the middle length is non-zero).

The third case is:

diagram

If the middle length is λ, then

$$\tau(u) + \tau(v) = 2\ell(g) + 2\lambda \geq 2\ell(g)$$

□

Subgroups of free groups

Proposition 3.0.4. If X is a (connected) graph, then $\pi_1(X)$ is free.

Proof when X has countably many cells. Let $T \subseteq X$ be a maximal tree, and let $\{e_1, e_2, \dots\}$ be the edges in X and not T. Let $X_N = T \cup \{e_1, \dots, e_N\}$. With this,

$$X = \bigcup_{n \geq 1} X_n$$

Pick a base vertex $v_0 \in T$. For each e_i, let α_i be the illustrated loop.

diagram

Lecture 10

Note

$$X_{n+1} = X_n \cup e_{n+1} = X_n \cup_{Y_{n+1}} (S^1 \cup I)$$

where Y_n is the following

diagram

By Seifert-van Kampen,

$$\pi_1(X_{n+1}) = \pi_1(X_n) * \langle \alpha_{n+1} \rangle$$

Thus, by induction, $\pi_1(X_n)$ is free for all n, and generated by $\alpha_1, \dots, \alpha_n$. When X is countably infinite, note that every (edge) loop $\gamma \subseteq X$ is contained in X_n for some n. Thus, $\pi_1(X)$ is generated by $\{\alpha_1, \alpha_2, \dots\}$.

By the universal property of free groups, we have a surjection

$$\eta : F_\infty = \langle \alpha_1, \dots \rangle \to \pi_1(X)$$

Suppose γ is a loop representing an element of $\ker(\eta)$. As before, γ is contained in X_n for some n. So γ is in the kernel of the map

$$\langle \alpha_1, \dots, \alpha_n \rangle \to \pi_1(X_n) \to \pi_1(X)$$

The first map is an isomorphism, so $\gamma \in \ker(\pi_1(X_n) \to \pi_1(X))$.

Since X_n is a retract[2] of X, every loop which is null-homotopic in X, is null-homotopic in X_n. So $\gamma = 1$ in $\pi_1(X_n) = \langle \alpha_1, \dots, \alpha_n \rangle \leq F_\infty$. □

Corollary 3.0.5. If G acts on a tree T freely, then G is free.

Proof. The action of G on T is a *covering space action*. Since T is simply connected, $X = G\backslash T$ is a graph, with universal cover T, and $G = \pi_1(X)$ is free. □

Corollary 3.0.6 (Nielsen-Schreier)**.** Any subgroup of $H \leq F_n$ is free.

Proof. Let $T = T_n$ be the Cayley tree of F_n. Then F_n acts on T freely, and so H acts freely on T. By the previous corollary, H is free. □

Remark 3.0.7. The choice of generating set comes from the choice of a maximal tree in the proposition.

[2] i.e. the inclusion $X_n \to X$ has a left inverse $r : X \to X_n$

4 Bass-Serre theory

We will study groups acting on trees, not necessarily freely. We will also see how to glue groups together, or cut groups into pieces.

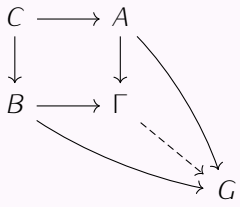
4.1 Amalgamated free products

Definition 4.1.1 (pushout)

A commutative diagram of groups

$$\begin{array}{ccc} C & \xrightarrow{i} & A \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle k} \\ B & \xrightarrow[\ell]{} & \Gamma \end{array}$$

is a *pushout* if for any group G, and homomorphisms $A \to G, B \to G$, there exists a unique homomorphism making the diagram



commute.

In this case, Γ is unique up to unique isomorphism, and therefore we may write $\Gamma = A \underset{C}{\sqcup} B$.

Theorem 4.1.2 (Seifert-van Kampen for cell complexes)**.** Suppose $K, L \subseteq X$ are subcomplexes, such that $X = K \cup L$. Suppose $K, L, K \cap L$ are all path connected. Then

$$\pi_1(X) = \pi_1(K) \underset{\pi_1(K \cap L)}{\sqcup} \pi_1(L)$$

Proof omitted. □

Note we use $\sqcup$ as it is a coproduct.

Proposition 4.1.3. Suppose $A = \langle S_A \mid R_A \rangle, B = \langle S_B \mid R_B \rangle$, $C = \langle \Sigma \mid \ldots \rangle$. Let i, j be represented by maps $\widehat{i} : \Sigma \to F(S_A), \widehat{j} : \Sigma \to F(S_B)$. Then

$$A \underset{C}{\sqcup} B = \left\langle S_A, S_B \mid R_A, R_B, \{\widehat{i}(\sigma)\widehat{j}(\sigma)^{-1} \mid \sigma \in \Sigma\} \right\rangle$$

Proof. Exercise. □

Example 4.1.4

If B is trivial, then

$$A \underset{C}{\sqcup} 1 = A / \langle\langle i(C) \rangle\rangle$$

Definition 4.1.5 ((amalgamated) free product)

If the maps i, j in the definition of a pushout are injective, then we write $\Gamma = A *_C B$, and call Γ the

amalgamated free product of A and B over C.

In particular, if $C = 1$, we write $\Gamma = A * B$, and we call this the *free product* of A and B.

Theorem 4.1.6 (Britton's lemma)**.** The *vertex group* A (or B) injects into $G = A *_C B$.

Remark 4.1.7. This is not true for pushouts. For example, $\mathbb{Z}/2 \coprod_{\mathbb{Z}} \mathbb{Z}/3 = 1$.

Lecture 11

To prove the theorem, we will construct a *graph of spaces* X, such that $G = \pi_1(X)$.

diagram

Let X_A be a presentation complex for A, and X_B be a presentation complex for B. As before, let Σ be a generating set for C. For each $\sigma \in \Sigma$, let α_σ be a based edge loop in X_A, representing $i(\sigma)$. Similarly, let β_σ be a based edge loop in X_B, representing $j(\sigma)$. To build this space:

1. Let X_A, X_B be the presentation complexes, with their based points.
2. Add in an edge t from the base point of X_A to the base point of X_B.
3. For each $\sigma \in \Sigma$, consider the following "rectangular" 2-cell **diagram** with gluing pattern $t\beta_\sigma^{-1}t^{-1}\alpha_\sigma$. Attach these to the diagram.

Call the resulting space X. By construction (and the Seifert-van Kampen theorem), $\pi_1(X) = G = A \coprod_C B$.

Proof. Suppose $g \in A$ mapsto $1 \in G = A *_C B$. Then g represented by a (based) loop γ in X_A, which is null-homotopic in X.

By van Kampen's lemma[3], γ bounds a singular disc diagram $D \to X$. Because the edge t appears in each rectabgle, and nowhere else, the rectangular 2-cells in D are arranged in strips, which we call t-corridors.

diagram

Since the boundary word is γ, which is contained in X_A. Therefore, we can't have any t on the boundary, so all of the t-corridors are annuli. Look at an *inner most disc* D_0 bounded by a t-corridor.

diagram

Since D_0 is contained in a t-corridor, it is contained in X_A or X_B. Without loss of generality (proof is symmetric), $D_0 \subseteq X_A$. Going around the t-corridor, we get a cyclic word δ in $\Sigma \cup \Sigma^{-1}$. In particular, $i(\delta)$ is the inner loop, $j(\delta)$ is the outer loop. But $i(\delta)$ bounds a disc D_0, and so it is contractible. So $i(\delta) = 1$. But i is injective, so $\delta = 1$. So $j(\delta) = 1$ in B.

By van Kampen's lemma, $j(\delta)$ has a van Kampen diagram $D_B \to X_B$. In particular, this has no t-corridors, and the same boundary as D_0 with its surrounding t-corridor. So we can remove D_0 and its surrounding t-corridor, and replace it with D_B.

diagram

This is now a van Kampen diagram, with one less t-corridor. Iterating, we can remove all of the t-corridors. But then we obtain a disc diagram Δ for γ with cells in X_A only. So $\Delta \to X_A$, and so $\gamma = 1$ in $\pi_1(X_A) = A$. □

Example 4.1.8

For a closed orientable surface Σ, we can cut along a curve γ to get

$$\pi_1(\Sigma) = \pi_1(\Sigma_A) *_{\mathbb{Z}} \pi_1(\Sigma_B)$$

What happens if we cut along a non-separating curve?

[3] Yes this isn't a presentation complex, it still applies.